in the brain regulates the release of hormones from the hypothalamus.

In the next chapter, we'll consider how the cellular and biochemical mechanisms we have discussed contribute to nervous system function on the system level.

Mastering Biology **The Visual Brain: Neural Conduction and Synaptic Transmission**

CONCEPT CHECK 48.4

1. How is it possible for a particular neurotransmitter to produce opposite effects in different tissues?
2. Some pesticides inhibit acetylcholinesterase, the enzyme that breaks down acetylcholine. Explain how these toxins would affect EPSPs produced by acetylcholine.
3. **MAKE CONNECTIONS** Name one or more membrane activities that occur both in fertilization of an egg and in neurotransmission across a synapse (see Figure 47.3).

For suggested answers, see Appendix A.

48 Chapter Review

Go to **Mastering Biology** for Assignments, the eText, the Study Area, and Dynamic Study Modules.

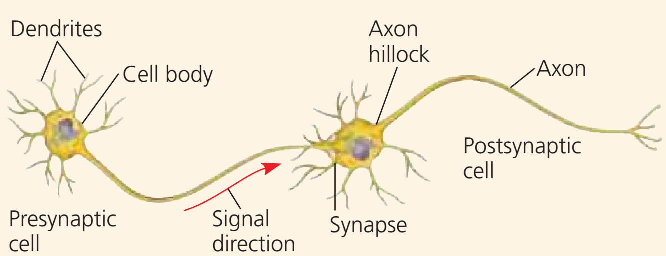
SUMMARY OF KEY CONCEPTS

To review key terms, go to the **Vocabulary Self-Quiz** in the **Mastering Biology** eText or Study Area, or go to goo.gl/zkjz9t.

CONCEPT 48.1

Neuron structure and organization reflect function in information transfer *(pp. 1068–1069)*

- Most **neurons** have branched **dendrites** that receive signals from other neurons and an **axon** that transmits signals to other cells at **synapses**. Neurons rely on **glia** for functions that include nourishment, insulation, and regulation.



- A **central nervous system (CNS)** and a **peripheral nervous system (PNS)** together process information in three stages: sensory input, integration, and motor output to effector cells.

? How would severing an axon affect the flow of information in a neuron?

CONCEPT 48.2

Ion pumps and ion channels establish the resting potential of a neuron *(pp. 1069–1072)*

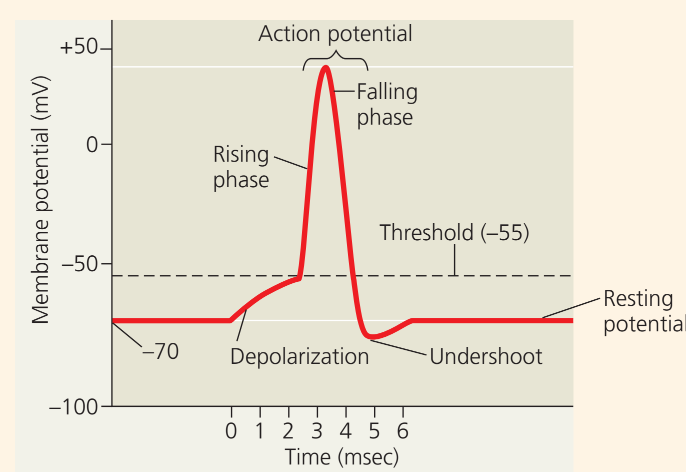
- Ion concentration gradients generate a voltage difference, or **membrane potential**, across the plasma membrane of cells. The concentration of Na^+ is higher outside than inside; the reverse is true for K^+. In resting neurons, the plasma membrane has many open potassium channels but few open sodium channels. Diffusion of ions, principally K^+, through channels generates a **resting potential**, with the inside more negative than the outside.

? Suppose you placed an isolated neuron in a solution similar to extracellular fluid and later transferred the neuron to a solution lacking any Na^+. What change would you expect in the resting potential?

CONCEPT 48.3

Action potentials are the signals conducted by axons *(pp. 1072–1077)*

- Neurons have **gated ion channels** that open or close in response to stimuli, leading to changes in the membrane potential. An increase in the magnitude of the membrane potential is a **hyperpolarization**; a decrease is a **depolarization**. Changes in membrane potential that vary continuously with the strength of a stimulus are known as **graded potentials**.
- An **action potential** is a brief, all-or-none depolarization of a neuron's plasma membrane. When a graded depolarization brings the membrane potential to **threshold**, many **voltage-gated ion channels** open, triggering an inflow of Na^+ that rapidly brings the membrane potential to a positive value. A negative membrane potential is restored by the inactivation of sodium channels and by the opening of many voltage-gated potassium channels, which increases K^+ outflow. A **refractory period** follows, corresponding to the interval when the sodium channels are inactivated.



- A nerve impulse travels from the axon hillock to the synaptic terminals by propagating a series of action potentials along the axon. The speed of conduction increases with the diameter of the axon and, in many vertebrate axons, with myelination. Action potentials in myelinated axons appear to jump from one **node of Ranvier** to the next, a process called **saltatory conduction**.

INTERPRET THE DATA *Assuming a refractory period equal in length to the action potential (see graph above), what is the maximum frequency per unit time at which a neuron could fire action potentials?*